

Mirror Media: Delivering timely and reliable news via the cloud

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By using the server architecture and tools of Google Cloud, Mirror Media is able to provide quality news stakeholders with timely and free access to more resources for development.

**Google
Cloud
results**

Reduced
software

ABOUT MIRROR MEDIA



About Mirror Media

Mirror Media is a Taiwanese news media company founded in 2016 running a print weekly magazine and an online news platform.

Publishing breaking news and feature stories on diverse topics from politics, finance, entertainment to culture, its written and image content have won several journalism awards.

Currently, its news site receives seven million monthly unique visitors.

Industries: Media & Entertainment

- Reduces operational workloads with serverless architecture empowered by Cloud Run
- Supports stable news services under traffic surges through Cloud Load Balancing
- Reduces DDoS attack frequency with Google Cloud Armor
- Enables quick integration of website abuse protection feature with reCAPTCHA Enterprise

deploy
from 1
minute



Location: Taiwan

Cloud Scheduler

iKala

About iKala Cloud

iKala Cloud is a Taiwanese digitalization and AI commerce solution provider founded in 2011. As a Google Cloud partner offering consultation services for cloud infrastructure, hybrid cloud solutions, machine learning and marketing applications, it helps enterprises overcome digital transformation challenges. The company has served more than 400 enterprise customers in Asia Pacific.

Google Cloud (<https://cloud.google.com/>)

Cloud Build (<https://cloud.google.com/build/>)

Cloud Load Balancing
(<https://cloud.google.com/load-balancing/>)

Cloud Run (<https://cloud.google.com/run/>)

Cloud Scheduler
(<https://cloud.google.com/scheduler/>)

and AI commerce solution provider
Cloud Storage (<https://cloud.google.com/storage/>)

Compute Engine
(<https://cloud.google.com/compute/>)

solutions, machine learning and marketing

As one of the leading news providers in Taiwan, **Mirror Media** (<https://www.mirrornews.net/>) is dedicated to delivering fast and accurate news. Running a print weekly magazine since 1978 and a news platform since 2016, the company distributes 100,000 magazine copies daily and welcomes seven million monthly visitors on its site.

Google Cloud Armor

(<https://cloud.google.com/armor/>)

company has served more than 100

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

Cloud Security Command Center

(<https://cloud.google.com/security-command-center/>)

Cloud Pub/Sub (<https://cloud.google.com/pubsub/>)

"Our primary mission is to bring quality news to readers as quickly as possible," says Hsin-Chan Chien, IT Director at Mirror Media. "Besides in-depth original content, we also partner with other media outlets to provide comprehensive coverage. Users can find all the important news on our website."

Offering quality online news in a timely manner means that Mirror Media needs a reliable IT infrastructure. Chien says that since breaking news frequently introduces traffic spikes, it would be more challenging and resource-consuming to support daily news services with an on-premise environment. As a result, the Mirror Media team decided to deploy its news platform in a public cloud from day one, opting for **Google Cloud** (<https://cloud.google.com/>) because of its user-friendly interfaces and cost-effective virtual machines (VMs).

"Based on our proven experiences, we believe that adopting a cloud environment is the most feasible way to provide daily news services despite traffic surges," he explains. "Google Cloud stood out among all the offerings with its intuitive interfaces and competitively priced computing resources."

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Reducing operational workloads with fully managed cloud services

Since its launch, Mirror Media has been running its online news services in Google Cloud. In the first few years, the Google Cloud team and its partner [iKala Cloud](https://ikala.cloud/en/) held meetings with the IT team in Mirror Media every quarter to discuss technical details of upcoming projects. Besides planning, iKala Cloud has also helped Mirror Media continuously optimize its services by providing security and feature reviews.

"The close collaboration with the Google Cloud team and iKala Cloud has greatly facilitated our adoption of the Google Cloud tools," notes Chien. "With their support, we've been able to drive continuous improvement of our products and operations."

Mirror Media first deployed the VMs in [Compute Engine](https://cloud.google.com/compute) as the servers of its news platform, gradually replacing them with Kubernetes containers in [Google Kubernetes Engine](https://cloud.google.com/gke) (GKE) for higher development and operational efficiency. In late 2022, the news provider decided to further reduce the operational workloads for its small IT team of 10 engineers by deploying its Kubernetes containers in [Cloud Run](https://cloud.google.com/run) and automating operations task scheduling with [Cloud Scheduler](https://cloud.google.com/scheduler). With the two fully managed tools, Mirror Media is able to redirect its resources towards new feature development.

"By supporting a serverless architecture, Cloud Run enables us to run Kubernetes containers with less operational effort. This is particularly beneficial for a small IT team like us," says Chien. "Before, one of our engineers had to spend one month changing the configuration of our containers whenever there was a Kubernetes update. With Cloud Run, we only need to check the infrastructure settings when we want to optimize our services, and our team can spend more time on product development."

On top of that, Mirror Media leverages Cloud Build (<https://cloud.google.com/build>) to automate its CI/CD workflow, which has enhanced its DevOps team's productivity. Before using Cloud Build, it needed two hours to deploy a new feature. With Cloud Build, the whole deployment process can be completed within 10 to 20 minutes.

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Enabling stable news services despite instant traffic surges

After breaking news or stories on trending topics are published, Mirror Media's news site frequently sees at least twice the amount of traffic spikes. With the autoscaling capability of Cloud Run and the traffic distribution feature of [Cloud Load Balancing](https://cloud.google.com/load-balancing?hl=en)

(<https://cloud.google.com/load-balancing?hl=en>) among containers, Mirror Media is able to provide highly available services with nearly zero downtime.

Through queuing messages with [Pub/Sub](https://cloud.google.com/pubsub) (<https://cloud.google.com/pubsub>) before data is stored in [Cloud Storage](https://cloud.google.com/storage) (<https://cloud.google.com/storage>), the team has ensured high availability of its database despite high volumes of incoming data.

"As a news provider, we prioritize the stability of our services, so that our readers can access the latest news anytime. With the highly scalable and available cloud infrastructure of Google Cloud, we're able to effortlessly keep our readers updated when something big happens," notes Chien.

Strengthening security to enhance service quality and credibility

Part of ensuring that its services remain highly available means enhancing security. The company's news site used to encounter distributed denial-of-service (DDoS) attacks once per year, leading to downtime of up to one hour. After the team deployed Google Cloud Armor (<https://cloud.google.com/armor>) to detect and mitigate cyber threats, it has almost never experienced any DDoS attacks. By using Security Command Center (<https://cloud.google.com/security-command-center?hl=en>) to regularly scan its cloud infrastructure, Mirror Media has also prevented security vulnerabilities in its services.

The quality of comment sections in online news outlets plays an important role in their credibility. To prevent abuse like robot comments, Mirror Media relies on reCAPTCHA to block spambots on its news web pages. It used to take the team two to three days to integrate reCAPTCHA with new features, but since reCAPTCHA Enterprise (<https://cloud.google.com/recaptcha-enterprise>) was made available in Google Cloud, Mirror Media takes advantage of the new service and reduces the time required for integration to several hours.

Expanding news features with data analytics and AI

Moving forward, Mirror Media plans to start analyzing user behavior data with BigQuery.

(<https://cloud.google.com/bigquery>) to provide more personalized services to its subscribers, such as artificial intelligence-powered content recommendation through Vertex AI (<https://cloud.google.com/vertex-ai>). To make its content more accessible, it will leverage Text-to-Speech AI (<https://cloud.google.com/text-to-speech?hl=en>) to convert written news into audio clips.

Chien says, "Since our foundation, Google Cloud has helped us build a popular and trustworthy news platform with high-performance infrastructure and security tools. Our next step is to leverage its data analytics and artificial intelligence capabilities to broaden our news services."

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